

**CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated  
for lumbosacral transitional anatomy**

[Author names removed for double-anonymized review]

*[Affiliations removed for double-anonymized review]*

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**Purpose:** A vertebra at the lumbosacral junction is named by counting. The gold standard counts caudally from C2 on whole-spine imaging<sup>1</sup>; however, a lumbar case is planned on a lumbar study<sup>2,3</sup>, T12 to S1, without C2. Abdominopelvic computed tomography (CT) holds that span plus the lowest ribs and pelvis, standing in for the preoperative view. Where a lumbosacral transitional vertebra (LSTV) alters the count, the local anatomy is ambiguous: four rib-free vertebrae may be an L1 with a lumbar rib or an L5 assimilated to the sacrum, and six may be a sixth lumbar vertebra, a T12 with aplastic ribs, or a lumbarized S1<sup>4</sup>. CTSpinoPelvic1K asks whether local morphology resolves that ambiguity without the count. Two public label sets already covered this imaging, CTSpine1K’s vertebrae and CTPelvic1K’s pelvis, on the same colonography patients, never joined; this release joins them on one series and annotates the bones neither had. It provides 802 CT records with per-level ribs and femora, lumbar levels anchored on the lowest rib-bearing vertebra and S1, and classes for L6, T13, a separate S1 and lumbar ribs, so anomalies are recorded as themselves.

**Acquisition and Validation Methods:** Records pair CTSpine1K<sup>5</sup> and CTPelvic1K<sup>6</sup> labels on each patient’s bone-richest acquisition under a VerSe-native scheme. Validation: geometric invariants (802/802 pass); rib–vertebra incidence across 5,749 evaluable ribs (0.035% offset); and spinopelvic measures matching published values.

**Data Format and Usage Notes:** NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a reference loader; archived at <https://doi.org/10.5281/zenodo.22139642>.

**Potential Applications:** Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar CT cohort, planning research (377 records prone). *Limitations:* thoracic ground truth is field-of-view limited; postural angles recumbent; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus.

## 6 I. INTRODUCTION

7 The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and  
8 a vertebra is identified by where it falls in that sequence. Transitional variants sit at the  
9 two ends of the lumbar column, the thoracolumbar transitional vertebra (TLTV) above and  
10 the lumbosacral transitional vertebra (LSTV) below, and they disturb identification in two  
11 ways at once. They change what the border vertebra and its ribs *look like*: a thoracolumbar  
12 border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral  
13 one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly  
14 wedged body, a reduced disc beneath it and hypoplastic facet joints, or conversely separated  
15 from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc<sup>4</sup>.  
16 They also change *how many* vertebrae each region contains: eleven or thirteen thoracic,  
17 four or six lumbar.

18 Either change alone would be tractable. Together they make it difficult to state defini-  
19 tively what any vertebra at these borders *is*. Four rib-free vertebrae above the sacrum may  
20 mean an L1 bearing a lumbar rib or an L5 assimilated to it; six may mean a true sixth  
21 lumbar vertebra, a T12 whose ribs are aplastic, or an S1 lumbarized away from the sacrum.  
22 Each set yields the same count, and the morphology that would separate the readings is not  
23 decisive to a human reader in every case. Whether it differs enough for a learned classifier  
24 to separate them, or at least to assign each reading a probability, is the question this release  
25 is built to support. Absent that, what remains is the count itself: the gold standard for  
26 numeration in transitional anatomy is whole-spine imaging, counting caudally from C2<sup>1</sup>,  
27 which fixes how many vertebrae and how many rib pairs the column holds, and no spine-  
28 limited acquisition can supply either count. LSTV is common, reported between 4% and  
29 30% depending on definition, and wrong-level surgery is its most serious consequence.

30 Existing public collections cannot address this, and size is not the reason (Table I). Spine  
31 datasets omit the pelvis; pelvic datasets do not number the vertebrae; TotalSegmentator,  
32 the whole-body scheme, has no class for a sixth lumbar vertebra or for a rib on a lumbar  
33 vertebra; and VerSe, the one collection with an L6 class, stops at the sacrum and carries  
34 no ribs. No public collection can therefore record a six-lumbar spine together with both  
35 of its borders, however good the segmentation, and a segmenter trained on them inherits  
36 the gap: faced with an enumeration anomaly it must shift the whole column by a level or

absorb a vertebra into its neighbor. The size of that problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS<sup>7</sup> raised that to 99%<sup>8</sup>. Its weights label L6 and T13 on CT, but its 1,536 training scans are in-house and unreleased: the capability exists as weights, not as data.

The intraoperative side of the problem has prior art of its own. LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT’s vertebral labels onto it, localizing the target level in a limited field of view to within a few percent of cases in clinical use<sup>9–11</sup>. The labels it projects are placed by hand on the CT, by a skilled operator and verified by the surgeon<sup>9</sup>: the method assumes a correctly labeled CT and does not supply one. Automating that step, and getting it right precisely where the anatomy is anomalous, is what a dataset carrying the classes those anomalies produce is for.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and gives a class to each structure an enumeration anomaly produces: a sixth lumbar vertebra, and a rib borne by a lumbar vertebra. A scheme without those classes lacks a class vocabulary expressive enough to describe anatomic anomalies.

The project began as a surgeon’s question: whether a vertebra can be named from the anatomy in front of the reader, on the images a lumbar case is actually planned on: by guideline those are lumbar studies<sup>2,3,12</sup>, T12 to S1, and whole-spine coverage is reserved for trauma with an identified injury<sup>13</sup> and for deformity radiographs. The cohort suits that question: an abdominopelvic CT runs from the diaphragm to the pelvic floor, so it holds the T12-to-S1 span of the planning study together with the lowest ribs and the whole pelvis, and it lacks C2 exactly as that study does. Every record comes from one prospective trial protocol, ACRIN 6664<sup>14,15</sup>: 802 patients aged 50 and over, scanned supine and prone on multidetector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol; the comparators pool several collections (CTSpine1K four, CTPelvic1K seven), are multi-site by design (VerSe), or are routine clinical scans under many protocols (TotalSegmentator). With the protocol held fixed, scanner effects on a model become measurable rather than confounded. The price is the field of view: abdominal, so only the lowest thoracic levels are present and C2 never is (Sec. V).

69 The gap was in plain sight. CTPelvic1K<sup>6</sup> and CTSpine1K<sup>5</sup>, released months apart by  
70 an overlapping author group, each annotated the COLONOG collection under radiologist  
71 supervision, one the pelvis and the other the vertebrae, on the same patients, and no one  
72 joined them, although the join needs no new imaging and no new radiologist. It is not a  
73 file merge: every patient was scanned prone and supine and neither source recorded which  
74 series it drew on, so each annotation had to be traced back to its own series (Sec. II.A).  
75 Done once, that crosswalk yields a spine-and-pelvis frame at 802 patients for the cost of  
76 the crosswalk, and the bones neither set had, ribs, femora, S1 and hardware, could then be  
77 annotated against it rather than from scratch. Two of the most-used public bone annotation  
78 sets were halves of one dataset; this release is the whole.

79 *Where the material is.* Everything described here is v8 of one archived dataset, <https://doi.org/10.5281/zenodo.22293878>; cite the concept DOI <https://doi.org/10.5281/zenodo.22139642>, which resolves to the current version (Sec. III).

## 82 I.A. Vertebra labeling when the scan cannot settle the count

83 *The limitation, stated plainly.* No scan in this corpus contains C2, so the conventional  
84 count cannot be performed: a property of abdominal imaging, not of the annotation. A  
85 thirteenth thoracic vertebra and a lumbar rib on L1 are the same object under two names,  
86 and either removes one rib-free vertebra from the count between the two borders of the  
87 lumbar spine, the last rib-bearing vertebra and the sacrum, whereas an aplastic twelfth rib  
88 adds one. A *stump rib* (a hypoplastic twelfth rib, a cardinal indicator of a thoracolumbar  
89 transitional vertebra) is still a rib: it leaves the count unchanged, and only its length records  
90 the anomaly. Without C2, a count of four therefore cannot separate a thirteenth thoracic  
91 vertebra from a lumbar rib, and a count of six cannot separate an aplastic twelfth rib from  
92 a sixth lumbar vertebra.

93 *What the release does in addition.* Every vertebra carries the identifier its radiologist-  
94 sourced annotation assigned, corrected where the source was wrong; those labels are the  
95 ground truth. Alongside them, every released measurement is also expressed against the  
96 two landmarks present in essentially every abdominal scan, the sacrum below and the lowest  
97 rib-bearing vertebra above, so that the measurement is positional rather than nominal: a  
98 vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its

TABLE I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it<sup>21</sup>.

| Collection                     | Scans | Count | L6 | Sacrum | S1 | Pelvis | Ribs | L. rib | Femora | Grade |
|--------------------------------|-------|-------|----|--------|----|--------|------|--------|--------|-------|
| CTSpine1K <sup>5</sup>         | 1,005 |       |    |        |    |        |      |        |        |       |
| CTPelvic1K <sup>6</sup>        | 1,184 |       |    |        |    |        |      |        |        |       |
| VerSe <sup>22</sup>            | 374   |       |    |        |    |        |      |        |        |       |
| RibSeg v2 <sup>23</sup>        | 660   |       |    |        |    |        |      |        |        |       |
| TotalSegmentator <sup>24</sup> | 1,204 |       |    |        |    |        |      |        |        |       |
| <b>CTSpinoPelvic1K</b>         | 802   |       |    |        |    |        |      |        |        |       |

length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these changes according to whether a reader would have said L5 or S1, so cases that disagree about the name remain comparable, and where the junction is in view the count is determined and the two descriptions agree.

*But the vertebrae themselves are known to differ.* Counting is not the only route to an identity. Level-specific morphometry is long established: pedicle width and height change systematically from the thoracic to the lumbar spine,<sup>16</sup> vertebral body, endplate and canal dimensions differ by level,<sup>17–19</sup> and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.<sup>4,20</sup> If those differences hold at the boundary itself, the phenotype is decidable from the vertebra in front of the reader without any global count. On this corpus it does hold: separating T12 from L1 on shape alone, with each patient’s own size divided out and cross-validation grouped by case, gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae. Size is removed because raw millimeters separate thoracic from lumbar trivially and teach nothing about the junction, where the two are nearly the same size.

That is reported as a property of the released measurements rather than as a method: the information needed to settle these variants is present locally.

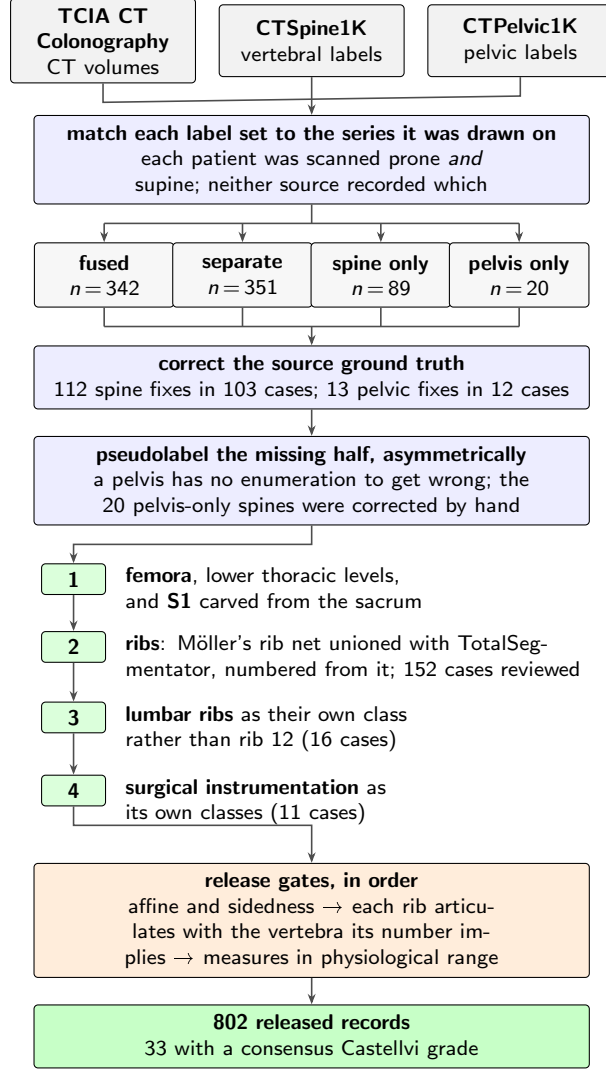


FIG. 1 How the dataset was built.  $n$  counts scans, or corrections, at each step.

## II. ACQUISITION AND VALIDATION METHODS

### II.A. Sources, and why a crosswalk was necessary

Both sources draw part of their imaging from the TCIA CT colonography collection<sup>14,15</sup>. CTSpine1K's 1,005 volumes come from four collections and CTPelvic1K's 1,184 from seven; COLONOG is the only collection common to both and the only one acquired under a single protocol, so this release is its COLONOG subset: 782 patients with a CTSpine1K vertebral annotation and 20 with a CTPelvic1K pelvic annotation only, 802 in all. Neither source released the mapping from an annotation to the specific CT series it was drawn on.

That mapping matters because every patient in CT COLONOGRAPHY was scanned

125 *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer.  
126 Turning a patient from supine to prone alters lumbar lordosis and the alignment of each  
127 segment against the next, so outlines drawn on one series are the wrong *shape* for the other  
128 and no rigid realignment recovers them. The crosswalk (Fig. 1) therefore resolves each  
129 annotation to a patient and then, within that patient’s volumes, to the series whose bone  
130 *agrees* with it: candidates are thresholded at bone attenuation and scored by overlap with  
131 the mask.

## 132 II.B. Fused and split records

133 CTSpine1K and CTPelvic1K each annotated a patient independently, so their labels are  
134 not guaranteed to come from the same CT series (Fig. 1). This turned out to matter for  
135 351 patients, nearly half the cohort: the two collections had labeled different acquisitions of  
136 the same person, one prone and one supine, with neither recording which. Because posture  
137 changes spinal alignment, a spine label from one acquisition cannot simply be paired with a  
138 pelvic label from the other. These 351 records are released but held out of any analysis that  
139 assumes a single posture; two labeled postures of one patient are a resource for studying  
140 alignment, not a defect.

## 141 II.C. Densification, and why it is asymmetric

142 A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing  
143 structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the  
144 safe direction: the sacrum and innominate bones carry no enumeration to get wrong, and  
145 quality is reported as held-out performance on the *pelvic\_only* records, withheld from training  
146 for that purpose. The reverse direction carries the whole problem this dataset addresses,  
147 since a pseudolabeled spine must commit to a count and on a transitional vertebra commits  
148 to the commoner reading; those 20 records were corrected by hand, tractable at that scale.

## 149 II.D. Ribs

150 No public rib annotation covers this cohort, and the two available automatic sources fail  
151 in complementary ways.



Möller’s binary rib network<sup>25</sup> reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented. On this corpus the segmentation itself failed on ribs that were *partially outside* the field of view: a rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. Its authors do not report this: their evaluation excluded subjects whose last rib was not visible, so it is recorded here as an observation from this cohort. In a colonography cohort the lower ribs are routinely clipped, and those are the ribs the count depends on.

TotalSegmentator<sup>24</sup> does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone: the most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib class labels, which matches each rib to the vertebra its number implies) has nothing to test.

The two were therefore unioned (Fig. 1), TotalSegmentator supplying the numbering and the medial-clipped cases, Möller the detection where its output is more complete.

The result is a pseudolabel whose review was triaged rather than exhaustive. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with, since a single bone carrying two numbers is a numbering error by construction, whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and four were not and are identified in the release. Reviewers were medical students working through a student research consortium<sup>26</sup>, trained against a written labeling protocol, with every correction recorded against the annotator who made it. Records passing the rule were not individually inspected, so the layer only guarantees freedom from the failure modes the rule detects, no more. The residual rib–vertebra offsets reported in Sec. II.G are the independent check on what the rule missed.

## II.E. Label scheme and counting anchors

The scheme is VerSe-native: vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure takes a fixed identifier above that range: S1 carved from the sacrum (29), hips (30–31),

182 femora (32–33), ribs (34–45 left, 46–57 right), lumbar ribs (58–59) and surgical hardware  
183 (60–66). The space is contiguous; there is no class for a thirteenth rib, because a rib on the  
184 vertebra below T12 is recorded as a lumbar rib whatever that vertebra is called.

185 Two anchors make the count decidable, and Fig. 2 shows both: the lowest rib-bearing  
186 vertebra above, **S1** carved from the sacrum below. Each earns its place by what it supports  
187 rather than by being an endpoint. S1 supplies the sacral endplate landmark that sacral  
188 slope and pelvic incidence are measured from, and with both brackets present L5 and L6  
189 are distinguishable where a spine-limited view alone would not be.

190 *How S1 is cut.* The sacrum’s outer boundary is the source annotation and is never altered;  
191 S1 is the cranial part of that same bone, separated by a plane. The plane’s normal is the  
192 sacrum’s own cranio-caudal axis, made orthogonal to a left–right axis measured directly  
193 rather than assumed, so as to remove the patient’s rotation within the scanner. That  
194 rotation is not negligible: median 4.5°, maximum 16.1°, and 508 of 801 records above 3°.  
195 The cut preserves the sub-division volume of the preceding release, changing the orientation  
196 of the S1/S2 boundary and not how much sacrum is called S1; per-record geometry ships  
197 with the archive.

198 Neither anchor is novel on its own: TotalSegmentator<sup>24</sup> carries a separate S1 class and  
199 per-level ribs, and VerSe<sup>22</sup> carries L6, whose identifier this release keeps, so its L6 *is* VerSe’s.

200 **A rib on a lumbar body receives its own class**, and this is the one class here with no  
201 published counterpart. A lumbar rib and a stump rib are the same object under two counts,  
202 so assigning it a number would commit the label to one reading of an enumeration anomaly.  
203 A scheme that numbers every rib 1–12 has nowhere to put a thirteenth: the annotator must  
204 either call it rib 12, asserting the vertebra beneath it is thoracic, the very question at issue,  
205 or discard it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe’s  
206 T13 is a vertebra rather than a rib. This scheme can record both: VerSe’s T13 (28) for a  
207 thoracic-type thirteenth vertebra with a true rib, and 58/59 for a lumbar-type body bearing  
208 a rudimentary one, so either reading is recorded as itself. No record here carries a T13; the  
209 sixteen lumbar-rib cases were read as lumbar-type bodies. Where a border vertebra could  
210 not be settled by its readers the label stands as read, and reclassifying those records with  
211 a classifier trained on local vertebral morphology (Sec. V) is the intended path for a later  
212 release.

213 Figure 2 shows why the interval is the primitive and the label is derived. Among the

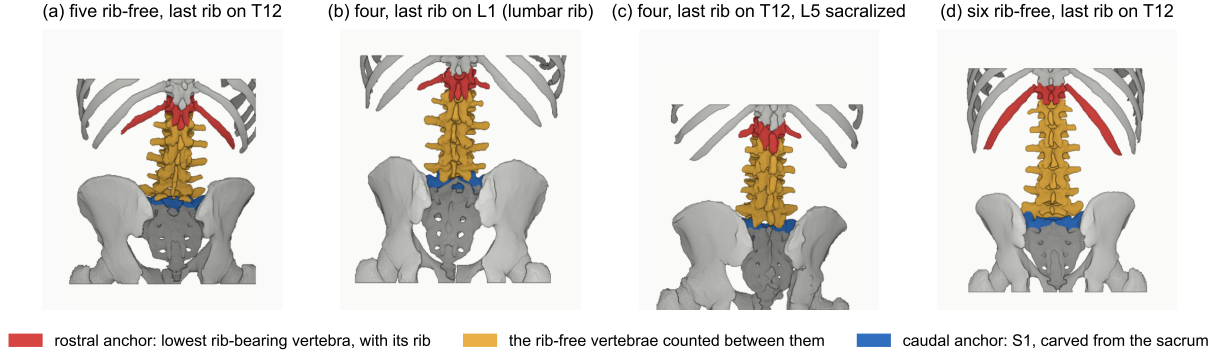


FIG. 2 The two anchors, from the released labels. Red, lowest rib-bearing vertebra and its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four: in (b) the last rib is on L1, in (c) on T12 with L5 assimilated to the sacrum.

records with four rib-free vertebrae, some reach that count from above, through a rib on L1, and some from below, through an L5 assimilated to the sacrum; the count alone cannot tell them apart, and the release can because the rib and the S1 boundary are both labeled.

Sixteen records carry a lumbar rib: thirteen bilateral and three unilateral, all on the right, a count too small to support any statement about side preference.

Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the eighteenth was not graded. The source transitional label, an independent field, reads lumbarization in fourteen, sacralization in two and semi-sacralization in one, and is unremarkable in the ungraded record: a spine can carry six lumbar-type vertebrae and still be read as a sacralization, depending on where the reader started the count.

Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`, `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by counting identifiers 25 and 58–59.

## II.F. Computational tools

*Access.* Every step is performed by open-source code archived with the dataset, together with the reference loader, the label scheme and the quality-control scripts that generated the tables in this manuscript.

*Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator<sup>24</sup> (`total` task, release 2.x) on a single graphics

processing unit. The binary rib network is Möller’s<sup>25</sup> released weights, applied through the nnU-Net v2 framework<sup>27</sup>. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble, applied out-of-fold: each record is completed by the fold that never trained on it, so no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (the repository URL identifies the authors; it is on the title page and will be restored here for the camera-ready version). Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image’s grid and affine exactly.

*Generative artificial intelligence.* A large language model (Claude, Anthropic) assisted with drafting and editing the manuscript text and with writing the analysis and figure code, all under author review. It was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes.

## II.G. Validation

Validation runs in three layers, and the geometric layer runs first: a measurement made on an inconsistent corpus returns a number that looks as finished as a correct one.

*Geometric invariants.* Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed. The check exists because a renumbering pass once wrote a reoriented array under a canonical affine and transposed a label away from its image, a failure invisible downstream.

The check compares *each sided pair separately*: testing the ribs alone passed all 802 records while four carried `left_hip` on the patient’s right, and pooling all sided structures into one test still missed one of those four, because a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

*Rib–vertebra incidence.* Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance;

263 of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is  
264 misnumbered. The denominator is stated because quoting the two offsets against all 11,548  
265 ribs would halve the rate by counting ribs the check never examined. Both offsets are  
266 field-of-view truncations on individually characterized cases, one at +1 level and one at -1,  
267 reported rather than suppressed.

268 The same check carries an invariant that reads as a null result. Of the 5,749 evaluable  
269 ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16  
270 records carry one, but because such a rib takes class 58 or 59 and never enters the numbered  
271 set the test examines. A numbered rib landing on a lumbar vertebra would be a counting  
272 error of exactly the kind this dataset exists to expose.

273 *Agreement with published values.* Derived spinopelvic measures are compared against  
274 Vialle *et al.*<sup>28</sup> (Table II). Pelvic incidence is the strongest of these checks because it is a  
275 morphological property of the pelvis rather than a posture, and so is directly comparable to  
276 a standing reference cohort.

## 277 II.H. Castellvi typing

278 Every record whose vertebral count departs from five carries a Castellvi grade<sup>29</sup> in the  
279 released manifest: 33 cases, typed I–IV with the *a/b* unilateral–bilateral qualifier. The grade  
280 and the count are *different axes*: grade IIIb occurs here at rib-free counts of four, five and  
281 six alike, and seven of the 33 graded cases carry a normal count of five.

282 Two radiology resident physicians graded every case independently and resolved their  
283 six disagreements by consensus; both reads and the consensus are in the manifest. The  
284 distinction the readers disagreed on, type II (articulation) against type III (bony fusion)<sup>4</sup>,  
285 is the one that matters clinically: four of the six disagreements were III against II.

## 286 II.I. Surgical hardware, and why a threshold is not enough

287 Metal is trivial to detect and hard to interpret, and the difference matters here because  
288 **an iatrogenic fusion is indistinguishable from a congenital one to a distance**  
289 **measurement**: a cage-bridged interspace reads as “no gap” exactly as a fused transitional  
290 vertebra does.

TABLE II Derived measures against a standing reference ( $n = 260$ )<sup>28</sup>; this cohort is supine.

| Measure                  | This dataset (supine) | Reference (standing) |
|--------------------------|-----------------------|----------------------|
| Pelvic incidence         | 54.7°                 | 54.7 ± 10.6          |
| Pelvic tilt              | 17.3°                 | 13 ± 6               |
| Sacral slope             | 36.9°                 | 41.0 ± 8.4           |
| Lumbar lordosis (supine) | 52.7°                 | 43 ± 11.2 (standing) |
| PI – LL mismatch         | 2.6°                  | centered near 0      |

291 *II.I.0.a. Detection over-calls by a factor of five.* A 2500 HU threshold inside a shell  
 292 around the labeled skeleton flags 52 of the 802 records. One of the two radiology resident  
 293 physicians read all 52 and confirmed 11, rejecting 41 (79%) as contrast, calcification and  
 294 reconstruction artefact. Attenuation does not separate the two groups (both saturate);  
 295 site and volume do, every confirmed implant at 2,586 mm<sup>3</sup> or above and every rejection at  
 296 1,768 mm<sup>3</sup> or below.

297 *II.I.0.b. The subtype block had to be extended.* Identifiers 60–63 name spinal instrumen-  
 298 tation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so  
 299 **64 arthroplasty**, **65 sacroiliac screw** and **66 osteosynthesis** were added: osteosynthesis  
 300 holds parts of one bone together where arthroplasty replaces a joint, and a shape rule calls  
 301 a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis, one  
 302 sacroiliac fixation and one pair of interbody cages (Fig. 3).

303 *II.I.0.c. Metal outranks bone, and that is a subtraction.* In every confirmed case the  
 304 implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than  
 305 adding them. In eight of these cases the femoral head that pelvic incidence and tilt are  
 306 measured from is an implant.

### 307 III. DATA FORMAT AND USAGE NOTES

308 Each record is a gzipped NifTI image/label pair sharing an identical  $4 \times 4$  affine, canon-  
 309 icalized to PIR, requiring no resampling. Masks are NifTI rather than DICOM, which is  
 310 what volumetric segmentation tooling expects.

311 Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the  
 312 data as `splits_5fold.json`. The five validation folds partition all 802 records, so the

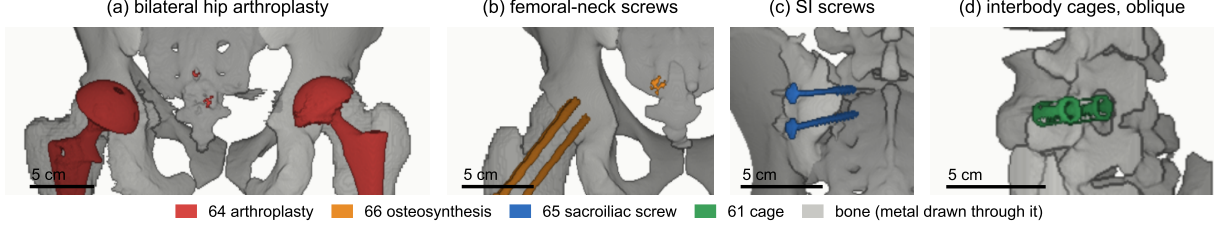


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 64,  $n = 8$ . (b) Femoral-neck screws, 66,  $n = 1$ . (c) Sacroiliac screws, 65,  $n = 1$ . (d) Interbody cages, 61,  $n = 1$ . Bar, 5 cm.

TABLE III Data types and metadata, and the records populating each, from the released manifest.

| Field                                                         | Records | Available |
|---------------------------------------------------------------|---------|-----------|
| Identifier, image path, label path, configuration, match type | 802     | 100.0%    |
| Patient position, tube potential, section thickness, kernel   | 802     | 100.0%    |
| Transitional label and class                                  | 802     | 100.0%    |
| Spine and pelvic annotation origin                            | 802     | 100.0%    |
| Instrumentation and partial-annotation flags                  | 802     | 100.0%    |
| Image/label alignment check                                   | 802     | 100.0%    |
| Spine series identifier; bone-fraction check                  | 782     | 97.5%     |
| Scanner manufacturer and model                                | 768     | 95.8%     |
| Sex                                                           | 749     | 93.4%     |
| Age                                                           | 709     | 88.4%     |
| Pelvic series identifier                                      | 713     | 88.9%     |
| Castellvi grade (consensus of two readers)                    | 33      | 4.1%      |

release provides cross-validation and *not* a held-out test set; users reporting a single held-out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag: every validation fold carries three sacralization and three lumbarization records. With 15 of each across 802 that is the most stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

Table III lists the key data types and metadata against the fraction of records for which

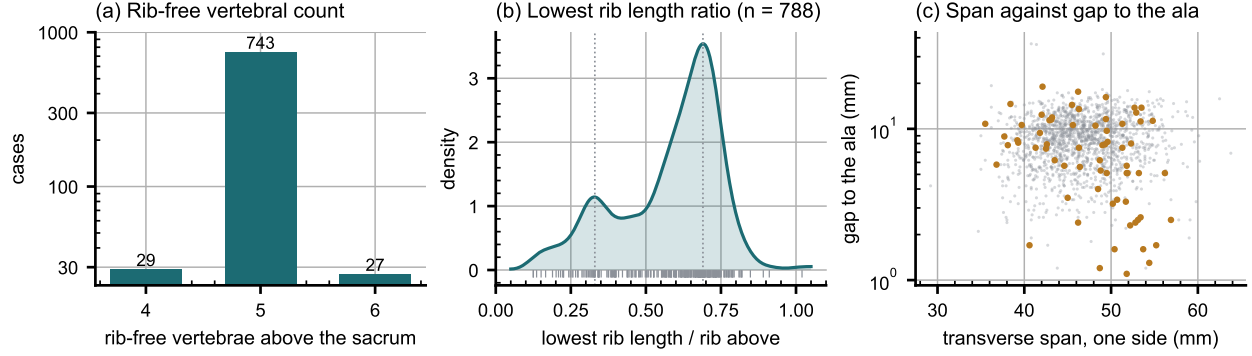


FIG. 4 Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b) Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to the ala; transitional labels marked.

each is populated.

The archive of record is Zenodo (<https://doi.org/10.5281/zenodo.22139642>). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

#### IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records. 738 are recorded female (393) or male (345); 11 carry DICOM’s *other* value and 53 no sex field. Age is present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven records outside the female/male dichotomy are excluded from sex-stratified measures rather than folded into “unrecorded”. The lower age bound is the screening guideline, not a selection applied here.

*Count-free measures* (Fig. 4). Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which of the two is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal.

*Level gradients and spinopelvic measures* (Fig. 5). Vertebral dimensions increase caudally; pelvic incidence does not vary with age while pelvic tilt does, the signature of a pelvis retroverting as a spine ages.

*Opportunistic measures*. Every scan carries a vertebral bone density measurement at no additional dose.<sup>30</sup> L1 trabecular attenuation is reported at the published standard site.



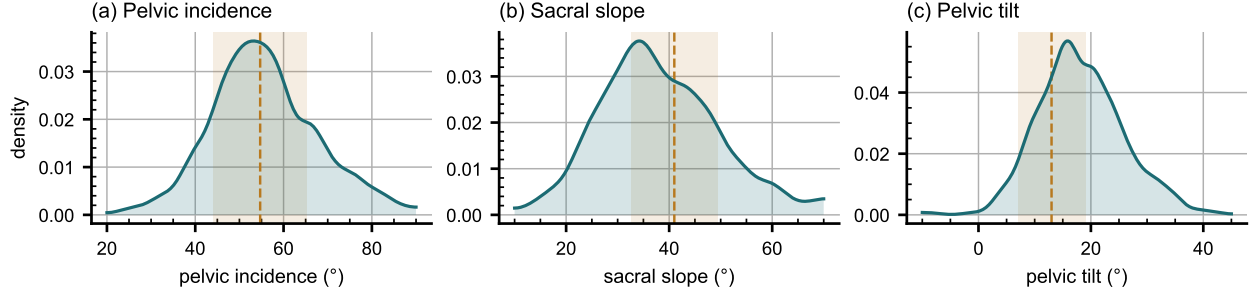


FIG. 5 Spinopelvic measures against standing reference bands (mean  $\pm$  SD)<sup>28</sup>.

## V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface*. Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, making sacral morphology measurable against the lumbar column rather than in isolation: the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads rather than from landmarks placed on a radiograph, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it: no trajectory can be planned across a junction whose segments cannot be named.

*A proxy for the preoperative view.* No public preoperative lumbar CT cohort exists that we know of, and VerSe, the largest public spine CT collection, lacks the sacrum and pelvis (Table I). This release is therefore the closest available stand-in for the view a lumbar case is planned on: the T12-to-S1 span with the lowest ribs and the pelvis; 377 of 802 records acquired prone, the position of posterior lumbar surgery; and classes for the anomalies that make level identification fail. It is a screening population at colonography dose, reconstructed with standard soft-tissue kernels rather than the bone kernel of a surgical CT, and not a surgical series: it supports method development for level identification and instrumentation geometry, not validation of a planning decision.

### V.A. Three uses the released measurements support

*Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and the cause most often cited is the transitional anatomy this co-

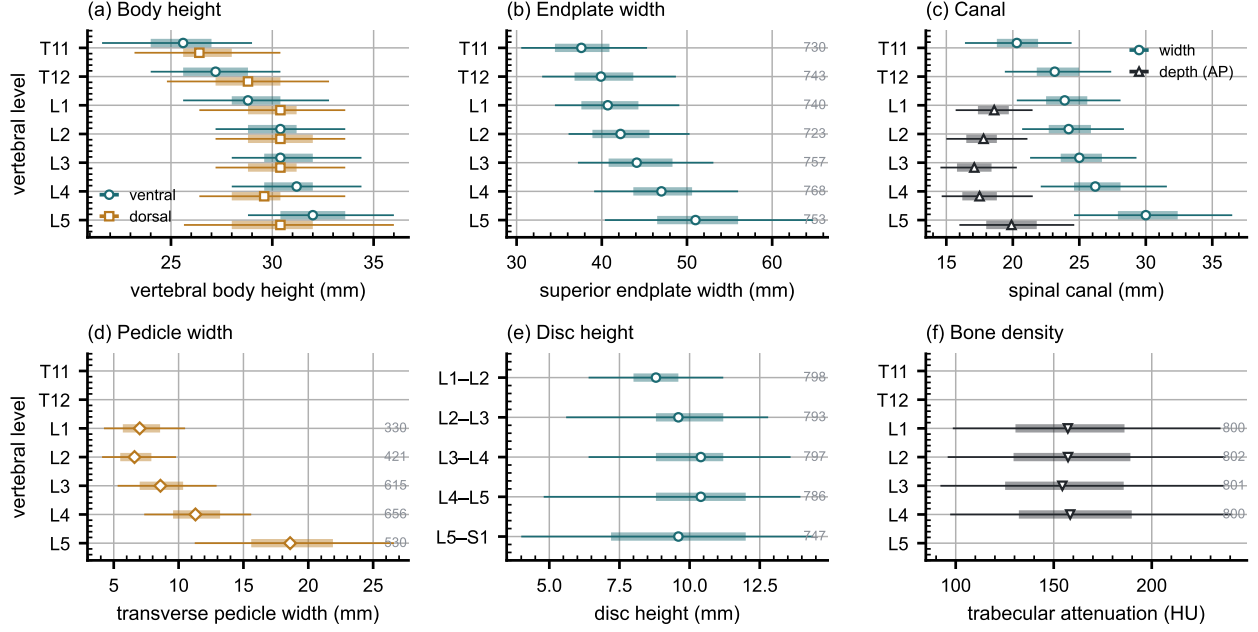


FIG. 6 Morphometry by level: median, interquartile range, 5th–95th percentile,  $n$  at right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height. (f) Trabecular attenuation.

hort was assembled around.<sup>31</sup> Per-level body height, canal width, endplate width, transverse  
process span and wedge ratio are released for T11–L5 across all 802 records, and because  
the labels were assigned against the twelfth rib rather than by enumeration, a model trained  
on them is not learning the convention that fails.

*Reference morphometry with its spread.* The values a surgeon consults for vertebral  
body height, canal dimensions and pedicle width come from cadaveric series of a few dozen  
specimens, plotted as single values with no indication of spread.<sup>19</sup> They cannot tell a reader  
whether the patient in front of them is unusual, because they never show what usual looks  
like. The same measures are given here from 802 records with the distribution attached  
(Fig. 6), reproducing two properties of the classical figures (body height crosses over, dorsal  
exceeding ventral at T11–T12 and reversing by L4–L5, and pedicle width widens steeply  
into the lower lumbar spine) and adding the width of each distribution, which is the part  
needed to judge an individual.

*Patient-specific models.* Planning is moving toward patient-specific biomechanical models  
rather than population norms.<sup>32</sup> Such a model needs spine, pelvis and femoral heads in one  
frame, which is this release’s construction; in 351 patients two postures give a within-patient

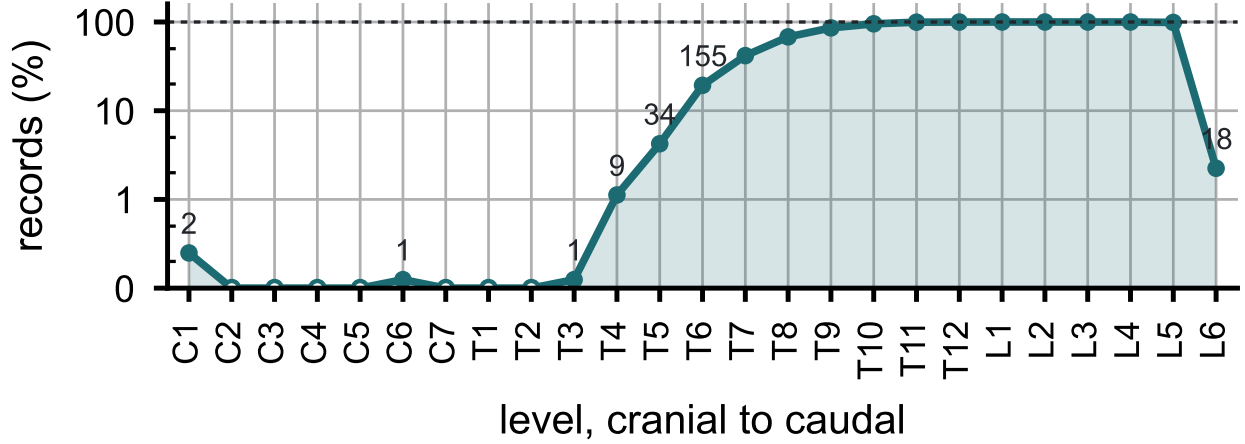


FIG. 7 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

change in alignment against which a predicted postural response can be checked.

Its limitations are stated at the level of detail a reader would need to catch them independently.

*Coverage.* Thoracic ground truth is field-of-view limited (Fig. 7) and does not extend to T1. Postural angles are supine; pelvic incidence is not postural and needs no such caveat. The cohort is a colorectal screening population aged 50 and over, so its distributions should not be read as representative of a surgical one.

*Strength of the labels varies by structure, and the release does not average over that.* Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong. Pelvic labels on records that lacked one are pseudolabeled. The rib layer is a pseudolabel whose human review was triaged by an automated rule rather than exhaustive, so its guarantee is the absence of the failure modes that rule detects. Transitional labels derive from two independent sources and are not uniformly adjudicated, which is precisely why the measures reported here are count-free, and the Castellvi grades are a consensus of two radiology resident physicians. S1 is carved from the sacrum by an automatic estimate of the S1–S2 boundary, and in 100 records that estimate is implausible: it makes S1 more than half the sacrum’s height or less than fifteen percent of it. Those records are flagged in the released quality-control table and should be excluded from any measurement that depends on the sacral endplate.

*Structures are not universally present.* Sacrum, hips and femora are present in all 802 records; one record carries no S1, one lacks T12, and nine lack an L5 identifier, because

the structure lies outside the field of view or, for L5, because the lowest lumbar segment is labeled L6 or incorporated into the sacrum. Users needing the caudal anchor should filter on it explicitly.

## VI. DISCUSSION

*Comparison with related material.* Against the collections in Table I, the contribution is the crosswalk placing spine and pelvis on the same series, together with classes for the anatomy an enumeration anomaly actually produces. VerSe<sup>21,22</sup> comes closest in intent, enriching for transitional anatomy and grading by Castellvi, but it does not segment the sacrum, so the structure a grade describes has no mask; this release segments it, as L6 or as a separately carved S1. Adding a sacrum to VerSe would not have supplied the pelvis, the femora or a prone acquisition, and here both label sets already existed: the join cost a crosswalk, not an annotation campaign. What this release adds over TotalSegmentator<sup>24</sup> is not more anatomy per scan but the classes an anomaly needs to be recorded accurately.

*Limitations.* The cohort comes from one acquisition protocol of a colorectal-screening population aged 50 and over, so how well it generalizes to a pathologic population is untested. The rib layer’s human review was triaged by an automated rule rather than performed on every bone, so it guarantees only the absence of the failure modes that rule detects.

*Future directions.* Imaging beyond this protocol, and the C2 anchor it cannot supply, are the natural next additions; applying the released weights to VerSe would add the S1 anchor and the anomaly classes to a bone-kernel spine collection, and VerSe’s vertebra labels would score that transfer.

## VII. CONCLUSION

CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of its own rather than recording it as something it is not. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2, which is every record here.

## 425 DATA AVAILABILITY

426 The release described here is v8, permanently archived at <https://doi.org/10.5281/zenodo.22293878>; the concept DOI <https://doi.org/10.5281/zenodo.22139642> resolves  
427 to the current version. The archive carries the loader, the label scheme, the quality-control  
428 tables and the analysis code under an open-source licence. The repository URL identifies  
429 the authors; it is on the title page and will be restored here for the camera-ready version.  
430

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## 434 CONFLICT OF INTEREST

435 The authors have no relevant conflicts of interest to disclose.

## 436 ETHICS

437 This work uses publicly available, de-identified imaging and annotations derived from it.  
438 The authors' institutional review board determined that it is not human participant research  
439 under 45 CFR 46 and requires no oversight.

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## 526 FIGURE CAPTIONS

527 **Figure 1.** How the dataset was built.  $n$  counts scans, or corrections, at each step.

528 **Figure 2.** The two anchors, from the released labels. Red, lowest rib-bearing vertebra and  
529 its rib; blue, S1; yellow, the rib-free vertebrae between. (b) and (c) both count four.

530 **Figure 3.** Instrumentation in the release, one exemplar each, drawn through bone. (a)  
531 Hip arthroplasty, 80,  $n = 8$ . (b) Femoral-neck screws, 82,  $n = 1$ . (c) Sacroiliac screws, 81,  
532  $n = 1$ . (d) Interbody cages, 77,  $n = 1$ . Bar, 5 cm.

533 **Figure 4.** Count-free measures. (a) Vertebrae between the lowest rib and the sacrum. (b)  
534 Lowest rib length over the rib above. (c) Lowest lumbar transverse span against its gap to  
535 the ala; transitional labels marked.

536 **Figure 5.** Spinopelvic measures against standing reference bands (mean  $\pm$  SD)<sup>28</sup>.

537 **Figure 6.** Morphometry by level: median, interquartile range, 5th–95th percentile,  $n$  at  
538 right. (a) Body height. (b) Endplate width. (c) Canal. (d) Pedicle width. (e) Disc height.  
539 (f) Trabecular attenuation.

540 **Figure 7.** Records carrying each vertebral level, cranial to caudal. Counts per identifier in



<sup>541</sup> Table S1.